

RecallGuard data flow

Revision: 2026-09-26.

1. Explicit memory write

API key -> authenticated owner -> consent=true -> input validation -> sensitive-pattern rejection -> exact duplicate lookup -> normalized embedding -> Chroma document/vector/metadata -> stored or duplicate response.

2. Chat

API key -> owner -> validate message and provenance reference -> sensitive-pattern rejection -> reject oversized base prompt -> optional extraction/admission if remember=true -> store decisions -> scoped retrieval/ranking -> bounded prompt -> Ollama -> response plus committed decisions and budget stats. An inference error returns generation_status=unavailable and preserves visibility of any preceding successful writes.

3. Retrieval

Authenticated owner -> query validation -> active owned IDs -> semantic candidates -> hybrid candidate admission -> ranking -> top-k. Expired and deleted rows never qualify.

4. Explicit correction

Authenticated owner -> owned active ID lookup -> validate new text/privacy -> replacement embedding -> update content/vector/source/version; retain original ID, creation time and expiry. Old content is not retained as a hidden version.

5. Delete / expiry

Owned DELETE -> Chroma delete -> subsequent ID lookup and retrieval cannot find the record.

Read-time expiry -> record hidden immediately.

Scheduled sweep -> physical expired-record deletion -> content-free counts.

6. Evaluation

Fixed synthetic JSONL -> isolated baseline and improved collections -> same encoder/query workload -> seven retrieval timings per case -> CSV, summary hash and error examples. Live generation is a separate labeled smoke check.